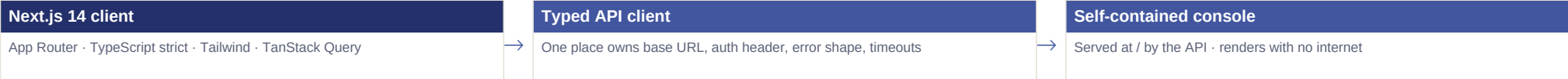


SYSTEM ARCHITECTURE — LAYERED VIEW

YatraAI is a three-tier application with a deliberately thick service layer. Business rules live in pure-Python domain objects that touch neither the database nor the web framework, which is what makes the optimiser, scorer and validator unit-testable in isolation and deterministic under a fixed seed.

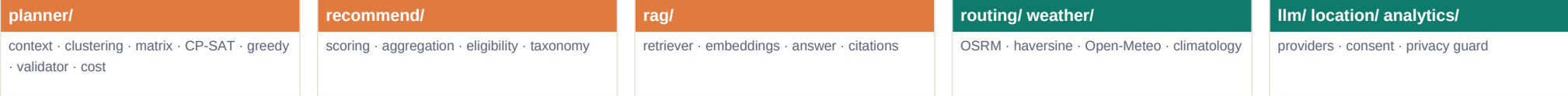
PRESENTATION



API



DOMAIN SERVICES — no framework or database imports



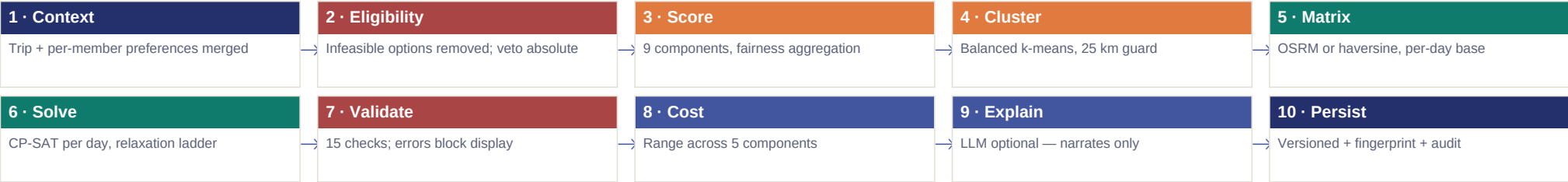
PERSISTENCE



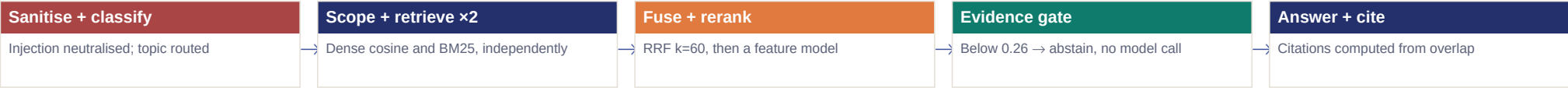
ARCHITECTURAL DECISION	RATIONALE	CONSEQUENCE
Domain logic in pure Python	The optimiser, scorer and validator import neither FastAPI nor SQLAlchemy.	Unit-testable without a database, and deterministic under a fixed seed.
Provider abstraction everywhere	LLM, embeddings, routing, weather and database are swappable configuration strings.	The product runs fully offline; each fallback is measured and disclosed, not hidden.
Validation independent of the solver	The validator re-reads persisted rows and re-derives every constraint from scratch.	A modelling bug surfaces as a failed validation, not as an impossible plan on screen.
Replanning as input change	All eleven actions re-run the whole pipeline; nothing edits a schedule in place.	A modification cannot produce a state the optimiser would never have generated.
Corpus derived from the catalogue	Chunks are generated from the same records the planner uses.	There is no second source of truth that can drift.

RUNTIME FLOWS

FLOW 1 — ITINERARY GENERATION · POST /api/v1/trips/{id}/itinerary



FLOW 2 — CITED ANSWER · POST /api/v1/assistant/ask



FLOW 3 — DATA INGESTION · make pipeline

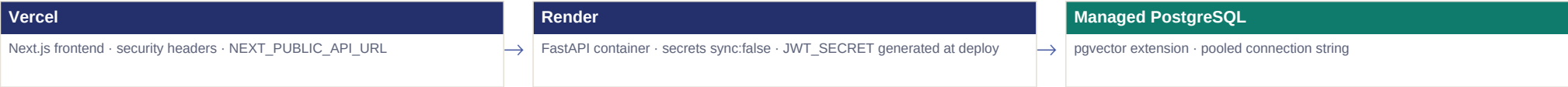


The same records also generate the knowledge corpus — 140 documents become 790 chunks and 790 embeddings — so the assistant and the planner can never disagree about a fact.

FAILURE MODE	SYSTEM RESPONSE
No feasible schedule for a day	Five-step relaxation ladder, least harmful first. Travel is relaxed last and least. What was relaxed is recorded and shown to the user; if all attempts fail the day returns empty with status INFEASIBLE.
Routing service unreachable	Haversine with documented detour factors and conservative urban speeds. Fallback rate reported at /metrics and disclosed in a UI banner.
Weather service unreachable	Committed per-cluster climatology, labelled as seasonal averages rather than a forecast.
Corpus cannot answer a question	Abstain below an absolute evidence threshold, with no model call and zero citations.
LLM provider errors or is absent	Deterministic template composition. The response records which provider actually answered.
A member's accessibility need conflicts with the majority	The candidate is removed, not down-weighted. A majority cannot outvote one member's mobility requirement.

DEPLOYMENT TOPOLOGY & DATA ARCHITECTURE

DEPLOYMENT



Local development and CI use Docker Compose (pgvector/pg16, API, web) or the SQLite fallback, which requires no infrastructure at all. GitHub Actions runs backend, Postgres integration, data-quality, frontend and docker jobs. scripts/verify_deployment.py exercises ten post-deployment checks against a live instance and exits non-zero on any failure.

DATA ARCHITECTURE — 31 TABLES, SIX DOMAINS



CROSS-CUTTING CONCERN	WHERE IT IS IMPLEMENTED	EVIDENCE IT WORKS
Determinism	num_workers = 1, fixed random seed, deterministic k-means seeding, input fingerprint stored per itinerary	A test asserts identical inputs produce identical plans, and that candidate ordering does not change the result
Portability across databases	GUID, JSONType, VectorType and TZDateTime TypeDecorators	The identical schema and the whole test suite run on PostgreSQL and on SQLite
Graceful degradation	Provider interfaces with measured fallbacks; nothing raises	/metrics reports per-provider latency, error rate and fallback rate; the UI shows an honest banner
Auditability	Append-only audit_logs on every sensitive trip action, carrying actor, entity, trip and request id	Rows join to structured logs by request id. Message bodies and coordinates are never included.
Privacy by construction	Coordinates snapped before the row is built; forbidden keys stripped at analytics write time; no history table exists	A test posts a real coordinate and asserts it appears nowhere in the dashboard payload
Reproducibility of claims	ml/ experiments, evaluation/ harness, generated reports	make experiments and python evaluation/run_rag_eval.py regenerate every published figure

The single architectural claim. Correctness is delegated to components that can be tested — a constraint solver, a validator, executable data contracts. Fluency is delegated to a component that cannot affect correctness, because it runs after everything has been decided and validated. Turn the language model off entirely, which is the default this test suite runs against, and the product still produces a complete, validated itinerary.